Mihir Jataniya

Software Engineer

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Summary

Software engineer with 1.5+ years of experience building production platforms end to end, from system design and real-time backends to cloud deployment. Specialised in applied AI, with hands-on work in RAG, LLM integrations and the reliability, latency and cost of AI systems in production. Comfortable working directly with customers and business stakeholders.

Experience

Full Stack Developer Feb. 2025 – Present

Glitch Over *Ahmedabad, India*

- Contributed to building a platform that **takes streamer and fan interaction to the next level** through **highly animated, real-time UIs**, serving **5,000+ daily active users**.
- Improved perceived speed based on **user feedback** on lag by **reducing network round trips (RTTs)**: **Redis caching**, **batched API calls**, **optimistic UI updates** and **CDN edge caching**, lowering API **p95 latency by 40%**.
- Made the backend **stateless and horizontally scalable**, using a **Redis adapter (pub/sub)** to fan out Socket.IO events across servers, keeping **1,000+ concurrent users** in sync with **sub-second updates**.
- Built a **customer-support bot** on **RAG with multi-source internal search** (vector + keyword, **MongoDB Atlas Vector Search**) that **auto-escalates to a human executive** on failure, **deflecting 70% of routine queries**.
- Designed **UI/UX** that makes the platform a richer **entertainment medium** for streamers and viewers, including **OBS overlays** that render **fan-triggered animated actions and alerts** live on stream.
- Built a **multi-provider payment system** on **Razorpay and Cashfree** that **auto-routes each transaction to the provider with the higher live success rate**, with **ACID guarantees** on all money movement.
- Designed **scalable NoSQL schemas on MongoDB** (embedding vs. referencing, indexes on hot paths) with **aggregation pipelines** powering **fast queries across all read paths**.
- Built separate **social-profile and gaming-profile factories**, each platform an adapter handling **auth, fetching and normalisation** (a new platform is **one adapter, zero core changes**), plus a **Discord** integration.

Projects

MeetNote AI | *MediaSoup, WebRTC, Socket.IO, AWS SQS, Deepgram, Gemini, React, Node.js* Live | GitHub

- Built a **full-stack real-time video-conferencing platform** with multi-party HD audio/video, live chat and AI-generated meeting notes; a single **SFU node** handles **50+ concurrent media streams**, scalable via configurable RTP port ranges.
- Engineered an **async recording, transcription and notes pipeline** decoupled through **AWS SQS** (chunked audio to Cloudinary, Deepgram, then LLM) on a separate worker; jobs **survive server restarts with zero loss**.
- Designed a **dual-LLM fallback chain** (Gemini primary, Groq/Llama secondary) with retry + backoff on 429/503, delivering **∼100% note-generation uptime** through provider rate limits.
- Implemented **JWT + Google OAuth**, a host-approval lobby, **ICS calendar export** and **PWA web-push**.
- Deployed on **AWS EC2** (nginx/TLS) with keyless **GitHub Actions + OIDC CI/CD**, deploying in **under 2 minutes**.

Multi-Provider LLM Client | *Python, asyncio, Pydantic, pytest, uv* GitHub

- **Async LLM client**: one interface over multiple providers via **pluggable adapters**; the reliability layer agents run on.
- **Cross-provider fallback** only on provider-unhealthy errors (429, 5xx, timeouts) while auth/validation errors fail fast; **exponential backoff with full jitter** honouring Retry-After within a **latency budget**; per-provider **circuit breakers**.
- Traced a benchmark's **5.2s p95** to silent SDK-internal 429 retries and disabled them so every attempt is counted and logged; **streaming with TTFT measurement** that retries only before the first token, so users never see duplicated text.
- **Structured output** via each provider's strict JSON-schema mode (Pydantic); **normalised token and USD cost accounting**; one **JSON log line per call** with caller metadata for **per-customer cost attribution**; 24 offline tests; **survived a live provider 503 outage** via retry + fallback.

Technical Skills

AI / LLM: LLM APIs, Tool Calling, Structured Output, RAG, Embeddings, Vector Search, Prompt Engineering, LLM Fallback, LangChain, LangGraph
Languages: TypeScript, JavaScript, Python, Java, SQL
Frontend / Backend: React, Next.js, Redux, Tailwind CSS, Node.js, Express, REST APIs, WebSockets, Socket.IO
Databases: MongoDB, PostgreSQL, Redis, pgvector
Infra & DevOps: AWS EC2, SQS, IAM, Amplify, Storage Buckets, Docker, GitHub Actions CI/CD, nginx, Load Balancing, Cloudflare CDN, DNS, TLS, Monitoring

Education

Mangalore Institute of Technology & Engineering Mangalore, India

B.E. in Computer Science *2021 – 2025*